

# Supplementary Information for

*Climate fluctuations shape primary-school academic performance through differential pathways: evidence from Côte d'Ivoire*

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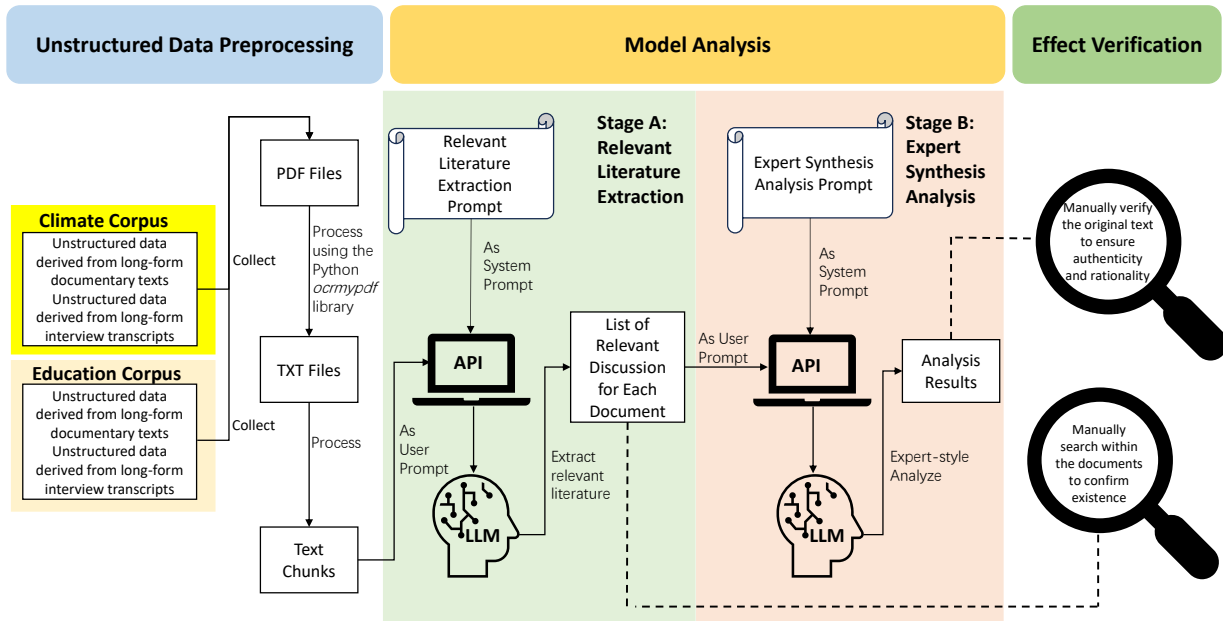
**Contents.** This document provides (i) the full prompts and reproducibility material for the LLM-based variable identification, (ii) the complete robustness regression tables (winsorisation, effective sample, finer-grained fixed effects, and re-estimated causal forests), (iii) a variable dictionary for the analytical variables and model symbols, and (iv) complementary methodological details that were not included in the main manuscript for length reasons.

**Project repository.** The accompanying project repository is available at <https://github.com/Andy-611/climate-education-civ>. It contains the associated materials used to support transparency and reproducibility.

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# S1 LLM-based variable identification: full prompts and rationale



**Figure S1. Workflow of the LLM-based variable-identification procedure.** The procedure begins with unstructured climate and education corpora, including long-form documentary texts and interview transcripts. These files are collected as PDF documents, converted into TXT files using the Python `ocrmypdf` library, and divided into text chunks for model input. In Stage A, a relevant-literature extraction prompt is passed to the LLM through the API as a system prompt, while the text chunks are supplied as user prompts; the model returns document-level excerpts relevant to predefined climate and education topics. In Stage B, the Stage A excerpts are supplied to a second LLM call under an expert-synthesis analysis prompt, producing written analytical conclusions about the appropriate climate indicator and household-economy proxy variables. The outputs are then manually verified against the original documents to confirm textual authenticity, source existence and inferential rationality.

## S1.1 Document corpora

We curated two corpora to anchor the model output:

**Climate corpus (24 documents).** Includes the Potsdam Institute for Climate Impact Research (PIK) *Climate Risk Profile: Côte d’Ivoire*<sup>1</sup>; the Britannica entry on Côte d’Ivoire’s climate<sup>2</sup>; the French-language Wikipedia entry<sup>3</sup>; peer-reviewed papers on West African monsoon and Intertropical Convergence Zone dynamics including Sultan et al.<sup>4</sup> and Niang et al.<sup>5</sup>; and additional grey-literature documents from the World Meteorological Organisation and Météo Côte d’Ivoire. All files are available in the project repository accompanying this study.

**Education corpus (28 documents).** Includes the UNESCO *Analyse du secteur de l’éducation*<sup>6</sup>, the UNESCO IICBA country brief<sup>7</sup>, the UNICEF report on education-aid cuts<sup>8</sup>, the MENA institutional pages<sup>9</sup>, the Chinese-language FOCAC overview of Côte d’Ivoire’s education system and the Ministry of Foreign Affairs (PRC) country snapshot. All files are available in the project repository accompanying this study.

All documents were OCR-ed with the `ocrmypdf` library, encoded as UTF-8, and chunked to 6,000-token segments that respect the DeepSeek-V3.2 context window<sup>10</sup>.

## S1.2 Two-stage chain-of-thought design

The model was queried in two stages. Stage A is a *fact-gathering* step that explicitly forbids any synthesis or translation; Stage B is an *expert-reasoning* step that triangulates the Stage A excerpts under role injection. This decomposition follows the chain-of-thought paradigm of Wei et al.<sup>11</sup> and the analyst-style prompting recommendations of Törnberg<sup>12</sup>.

**Stage A — climate corpus.**      System prompt:

*“Your task is to find, in the supplied text block, sentences that address any of the following topics: 1. Average level and variability of temperature, rainfall, wind speed and atmospheric pressure in Côte d’Ivoire; 2. How the Intertropical Convergence Zone, the West African Monsoon and the Harmattan wind influence Côte d’Ivoire’s climate; 3. Climate differences across regions of Côte d’Ivoire.*

*Return the excerpts verbatim, preserving the original language; never invent or translate. Output must be a Python list of strings. If no excerpt is found, return []. Do not output any other text.”*

**Stage B — climate corpus.**      System prompt:

*“You are a senior expert in climate science. Carefully analyse the supplied excerpts from documents on Côte d’Ivoire’s climate, and answer the question: ‘On the basis of the supplied material, compare the within-year variation of temperature, rainfall, wind speed and atmospheric pressure in Côte d’Ivoire. Given that Côte d’Ivoire straddles a tropical rainforest and a tropical savanna regime, which indicator(s) are most appropriate as the primary measure of seasonality and climate fluctuation?’*

*Requirements: (i) ensure factual and logical fidelity; (ii) after each conclusion, list the supporting excerpts in parentheses, identifying the file and exact quotation that justifies it.”*

**Stage A — education corpus.**      System prompt:

*“Your task is to find, in the supplied text block, sentences that address: 1. Fee structures and qualitative differences between public and private primary schools in Côte d’Ivoire; 2. Urban–rural disparities in Côte d’Ivoire in terms of economy, education and infrastructure.*

*Return the excerpts verbatim in their original language; never invent or translate. Output must be a Python list of strings. If no excerpt is found, return [].”*

**Stage B — education corpus.**      System prompt:

*“You are a senior expert in education and development economics. On the basis of the supplied excerpts from documents on Côte d’Ivoire, infer the average household economic conditions and gaps between (i) public vs. private primary-school families and (ii) urban vs. rural primary-school families.*

*Requirements: (i) ensure factual and logical fidelity; (ii) after each conclusion, list the supporting excerpts in parentheses, including file names and exact quotations.”*

### **S1.3 Summary of model conclusions and human verification**

The climate-stage B model output identifies rainfall as the binding climate dimension in Côte d’Ivoire. The southern coastal zone is under equatorial control of the Intertropical Convergence Zone (ITCZ), with two wet seasons (April–July and October–November) and annual totals of 1,500–2,500 mm; the north has a tropical-savanna regime with a single wet season (April–October) and 1,000–1,300 mm. Temperature, wind speed and pressure show modest intra-annual variation. Independent reading of all cited excerpts by the author confirmed factual accuracy.

The education-stage B output identifies school type (private vs. public) and school location (urban vs. rural) as the most credible proxies for household economic conditions: private schools are predominantly urban and serve more affluent families, whereas rural areas and public schools concentrate the poorest segments of the population. Cross-checking against UNESCO and UNICEF survey statistics confirmed this characterisation.

The intermediate outputs of the two-stage LLM procedure are provided in the project repository under the `llm-variable-identification-result/` directory. This directory contains two Stage A extraction files and two Stage B expert-synthesis reports.

## **S2 Robustness regression tables**

This section documents the four robustness checks summarised in the main text. All fixed-effects specifications include District and Date fixed effects, the Inverse Mills Ratio and grade dummies unless stated otherwise.

**Table S1.** Moderation model under winsorisation of the three rainfall indicators.

| Variable                    | 1% winsorisation |                | 5% winsorisation |                |
|-----------------------------|------------------|----------------|------------------|----------------|
|                             | PassRate         | ExcellenceRate | PassRate         | ExcellenceRate |
| Normr1h                     | −0.011***        | −0.018***      | −0.005*          | −0.012***      |
| Normr3h                     | 0.025***         | 0.031***       | 0.012***         | 0.022***       |
| NormFluctMonth              | −0.024***        | −0.017***      | −0.013***        | −0.010***      |
| SchType                     | 0.064***         | 0.040***       | 0.063***         | 0.041***       |
| SchLocation                 | −0.009***        | 0.019***       | −0.009***        | 0.020***       |
| Gender                      | −0.011***        | −0.009***      | −0.011***        | −0.009***      |
| IMR                         | −0.017***        | −0.010***      | −0.016***        | −0.009***      |
| Normr1h× SchType            | 0.007*           | −0.017***      | 0.002            | −0.019***      |
| Normr1h× SchLocation        | 0.014***         | 0.009***       | 0.010***         | 0.005*         |
| Normr1h× Gender             | −0.001           | −0.001         | −0.001           | −0.001         |
| Normr3h× SchType            | −0.001           | 0.016***       | 0.004            | 0.017***       |
| Normr3h× SchLocation        | −0.009***        | −0.011***      | −0.006**         | −0.009***      |
| Normr3h× Gender             | 0.000            | 0.001          | 0.000            | 0.001          |
| NormFluctMonth× SchType     | 0.005            | −0.025***      | 0.001            | −0.011*        |
| NormFluctMonth× SchLocation | 0.016***         | 0.008***       | 0.007***         | 0.001          |
| NormFluctMonth× Gender      | 0.001            | 0.001          | 0.001            | 0.000          |
| District FE                 | Yes              | Yes            | Yes              | Yes            |
| Date FE                     | Yes              | Yes            | Yes              | Yes            |
| N                           | 64,743           | 64,743         | 64,743           | 64,743         |
| $R^2$                       | 0.083            | 0.187          | 0.080            | 0.186          |

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . All specifications include grade dummies.

**Table S2.** LinearDML second-stage interaction estimates under winsorisation of the three rainfall indicators.

| Variable                         | 1% winsorisation     |                            | 5% winsorisation     |                            |
|----------------------------------|----------------------|----------------------------|----------------------|----------------------------|
|                                  | Pass $\tilde{R}$ ate | Excellence $\tilde{R}$ ate | Pass $\tilde{R}$ ate | Excellence $\tilde{R}$ ate |
| Norm $\tilde{r}$ 1h× SchType     | 0.007                | −0.026***                  | 0.006                | −0.025***                  |
| Norm $\tilde{r}$ 1h× SchLocation | 0.006*               | 0.000                      | 0.005                | −0.002                     |
| Norm $\tilde{r}$ 1h× Gender      | 0.000                | 0.000                      | 0.000                | 0.000                      |
| Norm $\tilde{r}$ 3h× SchType     | 0.004                | 0.020***                   | 0.007                | 0.019***                   |
| Norm $\tilde{r}$ 3h× SchLocation | −0.006**             | −0.008***                  | −0.005*              | −0.008***                  |
| Norm $\tilde{r}$ 3h× Gender      | 0.001                | 0.001                      | 0.001                | 0.001                      |
| NormFluctMonth× SchType          | 0.006                | −0.038***                  | 0.002                | −0.011*                    |
| NormFluctMonth× SchLocation      | 0.007**              | 0.005                      | 0.003                | 0.001                      |
| NormFluctMonth× Gender           | 0.001                | 0.002                      | 0.001                | 0.000                      |
| N                                | 64,743               | 64,743                     | 64,743               | 64,743                     |

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Tilded variables denote LassoCV-residualised quantities; entries are coefficients on the indicated residualised treatment-by-moderator terms.

**Table S3.** Moderation model on the effective-sample restriction ( $> 10$  and  $> 15$  participants).

| Variable                     | $> 10$ participants |                | $> 15$ participants |                |
|------------------------------|---------------------|----------------|---------------------|----------------|
|                              | PassRate            | ExcellenceRate | PassRate            | ExcellenceRate |
| Normr1h                      | −0.011***           | −0.017***      | −0.012***           | −0.019***      |
| Normr3h                      | 0.023***            | 0.026***       | 0.021***            | 0.028***       |
| NormFluctMonth               | −0.023***           | −0.017***      | −0.023***           | −0.018***      |
| SchType                      | 0.057***            | 0.045***       | 0.060***            | 0.051***       |
| SchLocation                  | −0.004*             | 0.021***       | −0.002              | 0.023***       |
| Gender                       | −0.013***           | −0.009***      | −0.015***           | −0.011***      |
| IMR                          | −0.015***           | −0.010***      | −0.013***           | −0.006**       |
| Normr1h × SchType            | 0.008*              | −0.020***      | 0.005               | −0.023***      |
| Normr1h × SchLocation        | 0.012***            | 0.008***       | 0.010***            | 0.009***       |
| Normr1h × Gender             | −0.001              | −0.001         | −0.000              | −0.000         |
| Normr3h × SchType            | −0.004              | 0.009          | −0.003              | 0.003          |
| Normr3h × SchLocation        | −0.009***           | −0.011***      | −0.010***           | −0.016***      |
| Normr3h × Gender             | 0.000               | 0.001          | −0.001              | 0.000          |
| NormFluctMonth × SchType     | 0.009               | −0.028***      | 0.006               | −0.033***      |
| NormFluctMonth × SchLocation | 0.016***            | 0.008***       | 0.018***            | 0.010***       |
| NormFluctMonth × Gender      | 0.001               | −0.000         | 0.001               | 0.000          |
| District FE                  | Yes                 | Yes            | Yes                 | Yes            |
| Date FE                      | Yes                 | Yes            | Yes                 | Yes            |
| N                            | 53,169              | 53,169         | 42,019              | 42,019         |
| $R^2$                        | 0.073               | 0.181          | 0.068               | 0.172          |

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .**Table S4.** LinearDML second-stage interaction estimates on the effective-sample restrictions.

| Variable                     | $> 10$ participants |                | $> 15$ participants |                |
|------------------------------|---------------------|----------------|---------------------|----------------|
|                              | PassRate            | ExcellenceRate | PassRate            | ExcellenceRate |
| Normr1h × SchType            | 0.008               | −0.027***      | 0.001               | −0.037***      |
| Normr1h × SchLocation        | 0.004               | −0.001         | 0.002               | −0.001         |
| Normr1h × Gender             | −0.000              | −0.001         | 0.000               | 0.001          |
| Normr3h × SchType            | 0.004               | 0.018**        | 0.007               | 0.014*         |
| Normr3h × SchLocation        | −0.007**            | −0.009***      | −0.008**            | −0.016***      |
| Normr3h × Gender             | 0.001               | 0.001          | 0.000               | 0.002          |
| NormFluctMonth × SchType     | 0.003               | −0.041***      | −0.005              | −0.046***      |
| NormFluctMonth × SchLocation | 0.006*              | 0.005          | 0.007**             | 0.008**        |
| NormFluctMonth × Gender      | 0.000               | 0.001          | −0.001              | −0.000         |
| N                            | 53,169              | 53,169         | 42,019              | 42,019         |

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Entries are coefficients on the indicated residualised treatment-by-moderator terms.

**Table S5.** Moderation model with finer-grained Region fixed effects (instead of District).

| Variable                     | PassRate  | ExcellenceRate |
|------------------------------|-----------|----------------|
| Normr1h                      | −0.013*** | −0.019***      |
| Normr3h                      | 0.018***  | 0.027***       |
| NormFluctMonth               | −0.026*** | −0.016***      |
| SchType                      | 0.063***  | 0.038***       |
| SchLocation                  | −0.011*** | 0.018***       |
| Gender                       | −0.011*** | −0.009***      |
| IMR                          | −0.018*** | −0.007***      |
| Normr1h × SchType            | 0.007     | −0.017***      |
| Normr1h × SchLocation        | 0.013***  | 0.010***       |
| Normr1h × Gender             | −0.001    | −0.001         |
| Normr3h × SchType            | −0.000    | 0.017***       |
| Normr3h × SchLocation        | −0.010*** | −0.011***      |
| Normr3h × Gender             | 0.000     | 0.001          |
| NormFluctMonth × SchType     | 0.006     | −0.026***      |
| NormFluctMonth × SchLocation | 0.016***  | 0.009***       |
| NormFluctMonth × Gender      | 0.001     | 0.001          |
| Region FE                    | Yes       | Yes            |
| Date FE                      | Yes       | Yes            |
| N                            | 64,743    | 64,743         |
| $R^2$                        | 0.087     | 0.189          |

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

## S2.1 Causal forest robustness

We re-estimated each of the six causal forests on (i) the 1% and 5% winsorised samples, and (ii) the  $> 10$  and  $> 15$  effective-participants samples. In both cases the qualitative pattern in Fig. 5 of the main text is preserved: public–rural pupils carry the largest negative effect of short-term and variable rainfall on the pass rate, and private pupils carry the largest negative effect on the excellence rate. Table S6 reports the preferred per-group CATEs; the corresponding robustness CATE exports are provided in the repository folder **robustnessCATEexports**, with separate files for the 1% winsorised, 5% winsorised,  $> 10$  effective-participants and  $> 15$  effective-participants specifications.<sup>1</sup>

<sup>1</sup>The four files are `Causal_Forest_Winsor_1pct_Group_Effects.xlsx`, `Causal_Forest_Winsor_5pct_Group_Effects.xlsx`, `Causal_Forest_EffectiveSample_gt10_Group_Effects.xlsx` and `Causal_Forest_EffectiveSample_gt15_Group_Effects.xlsx`.

**Table S6.** Per-group CATE estimates from the preferred causal forest (with sample share and  $p$ -values).

| Outcome        | Treatment      | Group                | Sample share | $\widehat{\text{CATE}}$ | $p$ -value            |
|----------------|----------------|----------------------|--------------|-------------------------|-----------------------|
| PassRate       | Normr1h        | Public-Rural-Female  | 28.67%       | -0.0151                 | 0.0064                |
|                |                | Public-Rural-Male    | 29.86%       | -0.0189                 | 0.0022                |
|                |                | Public-Urban-Female  | 17.58%       | -0.0040                 | 0.5409                |
|                |                | Public-Urban-Male    | 17.57%       | -0.0069                 | 0.3039                |
|                |                | Private-Rural-Female | 0.38%        | 0.0130                  | 0.2090                |
|                |                | Private-Rural-Male   | 0.48%        | 0.0199                  | 0.0773                |
|                |                | Private-Urban-Female | 2.74%        | 0.0167                  | 0.0787                |
|                |                | Private-Urban-Male   | 2.72%        | 0.0252                  | 0.0059                |
|                | Normr3h        | Public-Rural-Female  | 28.67%       | 0.0165                  | 0.0233                |
|                |                | Public-Rural-Male    | 29.86%       | 0.0186                  | 0.0213                |
|                |                | Public-Urban-Female  | 17.58%       | 0.0147                  | 0.0785                |
|                |                | Public-Urban-Male    | 17.57%       | 0.0205                  | 0.0223                |
|                |                | Private-Rural-Female | 0.38%        | 0.0002                  | 0.9872                |
|                |                | Private-Rural-Male   | 0.48%        | -0.0192                 | 0.1993                |
|                |                | Private-Urban-Female | 2.74%        | -0.0034                 | 0.7857                |
|                |                | Private-Urban-Male   | 2.72%        | -0.0226                 | 0.1135                |
|                | NormFluctMonth | Public-Rural-Female  | 28.67%       | -0.0177                 | $1.83 \times 10^{-7}$ |
|                |                | Public-Rural-Male    | 29.86%       | -0.0197                 | $1.13 \times 10^{-9}$ |
|                |                | Public-Urban-Female  | 17.58%       | -0.0145                 | 0.0040                |
|                |                | Public-Urban-Male    | 17.57%       | -0.0156                 | 0.0033                |
|                |                | Private-Rural-Female | 0.38%        | -0.0124                 | 0.0368                |
|                |                | Private-Rural-Male   | 0.48%        | -0.0123                 | 0.0801                |
|                |                | Private-Urban-Female | 2.74%        | -0.0012                 | 0.9309                |
|                |                | Private-Urban-Male   | 2.72%        | 0.0054                  | 0.6791                |
| ExcellenceRate | Normr1h        | Public-Rural-Female  | 28.67%       | -0.0209                 | 0.0005                |
|                |                | Public-Rural-Male    | 29.86%       | -0.0215                 | 0.0004                |
|                |                | Public-Urban-Female  | 17.58%       | -0.0061                 | 0.4389                |
|                |                | Public-Urban-Male    | 17.57%       | -0.0087                 | 0.2820                |
|                |                | Private-Rural-Female | 0.38%        | -0.0349                 | 0.0010                |
|                |                | Private-Rural-Male   | 0.48%        | -0.0278                 | 0.0089                |
|                |                | Private-Urban-Female | 2.74%        | -0.0411                 | 0.0004                |
|                |                | Private-Urban-Male   | 2.72%        | -0.0347                 | 0.0019                |
|                | Normr3h        | Public-Rural-Female  | 28.67%       | 0.0187                  | 0.0164                |
|                |                | Public-Rural-Male    | 29.86%       | 0.0172                  | 0.0423                |
|                |                | Public-Urban-Female  | 17.58%       | 0.0088                  | 0.3449                |
|                |                | Public-Urban-Male    | 17.57%       | 0.0230                  | 0.0108                |
|                |                | Private-Rural-Female | 0.38%        | 0.0459                  | 0.0031                |
|                |                | Private-Rural-Male   | 0.48%        | 0.0252                  | 0.0981                |
|                |                | Private-Urban-Female | 2.74%        | 0.0597                  | 0.0002                |
|                |                | Private-Urban-Male   | 2.72%        | 0.0364                  | 0.0310                |
|                | NormFluctMonth | Public-Rural-Female  | 28.67%       | -0.0163                 | $6.56 \times 10^{-6}$ |
|                |                | Public-Rural-Male    | 29.86%       | -0.0175                 | $2.05 \times 10^{-7}$ |
|                |                | Public-Urban-Female  | 17.58%       | -0.0071                 | 0.1858                |
|                |                | Public-Urban-Male    | 17.57%       | -0.0083                 | 0.1467                |
|                |                | Private-Rural-Female | 0.38%        | -0.0532                 | 0.0159                |
|                |                | Private-Rural-Male   | 0.48%        | -0.0396                 | 0.0153                |
|                |                | Private-Urban-Female | 2.74%        | -0.0743                 | $2.35 \times 10^{-5}$ |
|                |                | Private-Urban-Male   | 2.72%        | -0.0564                 | 0.0003                |

$\widehat{\text{CATE}}$  denotes the conditional average treatment effect at the indicated combination of (SchType, SchLocation, Gender).



### S3 Variable dictionary

Table S7 defines the analytical variables and model symbols used in the main manuscript, Extended Data and Supplementary Information.

**Table S7.** Variable dictionary.

| Variable                                   | Definition                                                                                                                                                                              |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $i$                                        | Index for an analytical observation. In the outcome models, an observation is a school-round-gender-grade cell; in the Heckman first stage, $i$ indexes schools in the selection frame. |
| $t$                                        | Date or school-round index. Because rainfall is available at ten-day frequency, $t-3$ denotes the previous month in the construction of <i>FluctMonth</i> .                             |
| $g$                                        | Grade-level index. Five grade dummies, $G_{i,g}$ , represent the six primary-grade categories.                                                                                          |
| $N$                                        | Number of pupils in the relevant school-round-gender-grade cell.                                                                                                                        |
| $n_{0 \text{ err}}$                        | Number of pupils with zero errors in the Daily Dictation test.                                                                                                                          |
| $n_{1-5 \text{ err}}$                      | Number of pupils with one to five errors in the Daily Dictation test.                                                                                                                   |
| $n_{\geq 6 \text{ err}}$                   | Number of pupils with six or more errors in the Daily Dictation test. These pupils are outside the pass-rate numerator.                                                                 |
| PassRate                                   | Share of pupils with zero to five errors.                                                                                                                                               |
| ExcellenceRate                             | Share of pupils with zero errors.                                                                                                                                                       |
| $Y_{i,t}$                                  | Generic educational outcome in the regression equations; either PassRate or ExcellenceRate.                                                                                             |
| $r1h_{i,t}$                                | One-month cumulative rainfall matched to observation $i$ at date $t$ , measured from the Department-level OCHA/CHIRPS rainfall data.                                                    |
| $r3h_{i,t}$                                | Three-month cumulative rainfall matched to observation $i$ at date $t$ , measured from the Department-level OCHA/CHIRPS rainfall data.                                                  |
| $r1h_{i,t-3}$                              | One-month cumulative rainfall for the same Department in the previous month.                                                                                                            |
| $\mu_{r1h}, \sigma_{r1h}$                  | Sample mean and standard deviation used to standardise $r1h$ .                                                                                                                          |
| $\mu_{r3h}, \sigma_{r3h}$                  | Sample mean and standard deviation used to standardise $r3h$ .                                                                                                                          |
| $\mu_{FluctMonth}, \sigma_{FluctMonth}$    | Sample mean and standard deviation used to standardise <i>FluctMonth</i> .                                                                                                              |
| $FluctMonth_{i,t}$                         | Relative month-to-month rainfall fluctuation, defined as $ r1h_{i,t} - r1h_{i,t-3}  / r1h_{i,t}$ .                                                                                      |
| Normr1h <sub><math>i,t</math></sub>        | Standardised one-month rainfall accumulation, interpreted as the short-term rainfall-level indicator.                                                                                   |
| Normr3h <sub><math>i,t</math></sub>        | Standardised three-month rainfall accumulation, interpreted as the longer-run rainfall-level indicator.                                                                                 |
| NormFluctMonth <sub><math>i,t</math></sub> | Standardised month-to-month rainfall fluctuation, interpreted as the rainfall-variability indicator.                                                                                    |
| Normr1h, NormFluctMonth                    | LassoCV-residualised versions of the three standardised rainfall treatments in the LinearDML specifications.                                                                            |
| SchType <sub><math>i</math></sub>          |                                                                                                                                                                                         |
| SchLocation <sub><math>i</math></sub>      | Binary school-type indicator; private schools are coded 1 and public schools are coded 0.                                                                                               |
| Gender <sub><math>i</math></sub>           | Binary school-location indicator; urban schools are coded 1 and rural schools are coded 0.                                                                                              |
| $G_{i,g}$                                  | Binary pupil-gender indicator; male pupils are coded 1 and female pupils are coded 0.                                                                                                   |
|                                            | Grade dummy variables included as controls in the regression models.                                                                                                                    |

| Variable                                   | Definition                                                                                                                                                                           |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Department code                            | Administrative boundary code used to link school observations with Department-level OCHA/CHIRPS rainfall records.                                                                    |
| DistrictFE <sub><i>i</i></sub>             | District fixed effects absorbing time-invariant district-level heterogeneity.                                                                                                        |
| DateFE <sub><i>t</i></sub>                 | Date fixed effects absorbing shocks common to all observations in the same school round/date.                                                                                        |
| RegionFE <sub><i>i</i></sub>               | Region fixed effects used in the robustness specification that replaces District fixed effects with finer-grained Region fixed effects.                                              |
| School-year dummies                        | Time controls used in the LinearDML nuisance models to absorb school-year-specific variation.                                                                                        |
| ElectrifiedSch <sub><i>i</i></sub>         | Binary school-infrastructure indicator for whether the school has electricity; used in the Heckman first-stage selection model.                                                      |
| MultimediaRoom <sub><i>i</i></sub>         | Binary school-infrastructure indicator for whether the school has a multimedia room; used in the Heckman first-stage selection model.                                                |
| InternetSch <sub><i>i</i></sub>            | Binary school-infrastructure indicator for whether the school has an internet connection; used in the Heckman first-stage selection model.                                           |
| ProjectDensity <sub><i>i</i></sub>         | Local participation-density exclusion restriction: the share of schools in the relevant IEPP cluster that participate in the Daily Dictation programme.                              |
| $D_i^*$                                    | Latent propensity for school <i>i</i> to participate in the programme and report usable Daily Dictation data.                                                                        |
| $D_i$                                      | Observed selection indicator equal to 1 when school <i>i</i> both participates and reports usable data, and 0 otherwise.                                                             |
| $z_i$                                      | Vector of first-stage Heckman selection covariates: intercept, school type, school location, infrastructure indicators and project participation density.                            |
| IMR <sub><i>i</i></sub>                    | Inverse Mills Ratio from the Heckman first-stage probit, added to outcome regressions to correct for non-random participation and reporting.                                         |
| $\mathcal{T}$                              | Set of rainfall treatments in the moderation model: {Normr1h, Normr3h, NormFluctMonth}.                                                                                              |
| $\mathcal{M}$                              | Set of moderators in the moderation model: {SchType, SchLocation, Gender}.                                                                                                           |
| $T_{i,t}$                                  | Focal rainfall treatment in the moderation, LinearDML and causal-forest models; one of Normr1h, Normr3h or NormFluctMonth.                                                           |
| $M_i$                                      | Focal moderator in the moderation model; one of SchType, SchLocation or Gender.                                                                                                      |
| $T_{i,t}M_i$                               | Treatment-by-moderator interaction term in the moderation model.                                                                                                                     |
| $X_i$                                      | Effect-modifier vector in the causal forest: (SchType <sub><i>i</i></sub> , SchLocation <sub><i>i</i></sub> , Gender <sub><i>i</i></sub> ).                                          |
| $W_i, \mathbf{W}$                          | Control or nuisance-variable vector used for residualisation; includes the Inverse Mills Ratio, the other two rainfall indicators, grade dummies, district dummies and time dummies. |
| $\tilde{Y}, \tilde{T}$                     | Residualised outcome and residualised treatment after removing variation explained by nuisance controls in LinearDML or CausalForestDML.                                             |
| $\hat{f}(\mathbf{W}), \hat{g}(\mathbf{W})$ | LassoCV nuisance functions used in LinearDML to predict the outcome and focal treatment from controls.                                                                               |
| $\hat{m}(X_i, W_i), \hat{e}(X_i, W_i)$     | CausalForestDML nuisance functions used to residualise the outcome and focal treatment before estimating heterogeneous treatment effects.                                            |
| $\tau(x)$                                  | Marginal conditional average treatment effect for target group <i>x</i> , defined as the derivative of the expected outcome with respect to the continuous rainfall treatment.       |
| $\hat{\tau}(x)$                            | Estimated conditional average treatment effect for target group <i>x</i> .                                                                                                           |

| Variable                               | Definition                                                                                                                        |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| $\widehat{\text{CATE}}$                | Reported group-level conditional average treatment-effect estimate.                                                               |
| $\alpha_i(x)$                          | Adaptive forest weight assigned by honest trees to observation $i$ when estimating the treatment effect for group $x$ .           |
| $\beta$ coefficients                   | Outcome-model coefficients in the OLS, fixed-effects, Heckman-corrected and moderation regressions.                               |
| $\gamma$ coefficients                  | Heckman first-stage probit coefficients.                                                                                          |
| $\rho_g$                               | Grade-dummy coefficient for grade level $g$ .                                                                                     |
| $\theta$ coefficients                  | Moderation or heterogeneous-treatment-effect coefficients, including treatment-by-moderator terms.                                |
| $\varepsilon_{i,t}, \nu_i, \eta_{i,t}$ | Error terms in the outcome regressions, Heckman selection equation and LinearDML second-stage equation, respectively.             |
| $\phi(\cdot), \Phi(\cdot)$             | Standard-normal probability density function and cumulative distribution function used to compute the Inverse Mills Ratio.        |
| $p$ -value                             | Probability value used to evaluate statistical significance in the regression and CATE tables.                                    |
| $R^2$ , adjusted $R^2$                 | Model-fit statistics reported for the linear regression specifications.                                                           |
| Sample share                           | Share of the analytical sample belonging to the subgroup shown in the group-level CATE table.                                     |
| Feature importance                     | Normalised causal-forest importance score indicating how much each effect modifier contributes to treatment-effect heterogeneity. |

## S4 Additional methodological notes

### S4.1 Why a causal forest is needed in addition to LinearDML

LinearDML preserves the interpretability of linear interaction coefficients while orthogonalising out high-dimensional confounding. However, it imposes a global linear functional form for both the main effect and the moderation. Wager and Athey<sup>13</sup> and Athey, Tibshirani and Wager<sup>14</sup> prove asymptotic normality of the causal forest treatment-effect surface under honest splitting and cross-fitting, which allows pointwise confidence intervals on heterogeneous CATEs without assuming a functional form. In our application the causal forest corroborates the LinearDML moderation estimates while exposing non-linear heterogeneity that LinearDML cannot capture, especially the disproportionately large negative excellence-rate effects in small private-school subgroups.

### S4.2 Choice of $r1h$ , $r3h$ and $FluctMonth$

The three indicators capture complementary dimensions of the rainfall regime.  $r1h$  captures the contemporaneous rainfall load on the school round;  $r3h$  captures the seasonal rainfall sum that drives subsistence agricultural income;  $FluctMonth$  captures relative month-to-month volatility. The one- and three-month totals are mechanically related ( $\rho = 0.494$ ; Extended Data Table ED2, the correlation matrix), but their variance-inflation factors remain below 2, and the variability indicator has lower correlations with both level measures ( $\rho = -0.248$  with Normr1h and  $\rho = 0.166$  with Normr3h). This supports treating the three variables as distinct empirical channels rather than interchangeable rainfall proxies.

### S4.3 Software and computational reproducibility

All numerical results were generated with the following stack: Python 3.12.10; `econml` 0.16.0; `scikit-learn` 1.8.0; `statsmodels` 0.14.6; Stata 19 (`reghdfe` and `xtreg`). LinearDML used random seed 120 and each causal forest used random seed 32 on a workstation with 16 logical CPUs.

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